

(Re)Shaping Energy Finance

A Systematic Literature Review

Anagnoste · Andrei · Bolovăneanu · Cepoi · Clodnitchi · Cramer · Grecu · Lessmann · Pele · Petukhina · Strat

Bucharest University of Economic Studies (ASE) · Humboldt-Universität zu Berlin · HTW Berlin

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Why This Review?

The intersection of AI and Energy Finance — a field in rapid transformation



Energy Transition

Global shift toward net zero is reshaping financial markets. Energy products are increasingly viewed as financial assets — intensifying the energy-finance nexus.



Geopolitical Shocks

Russia-Ukraine war amplified the global energy crisis. Regulatory responses, price caps and ESG mandates have further linked energy and financial markets.



AI Explosion Post-2020

Over 95% of AI-related studies in energy finance were published after 2020. COVID-19 and the energy crisis catalyzed a massive methodological shift.

The Research Gap

No prior study has done this systematically

> *"No existing review has systematically analyzed how AI techniques have been applied within the energy finance literature - nor whether they outperform classical econometric approaches across the six pillars."*

01 Breadth

Previous surveys acknowledged AI's role but never mapped it across all 6 energy finance pillars simultaneously.

02 Depth

No assessment of whether methodology fits the data - or whether AI genuinely outperforms classical econometrics in each domain.

03 Emerging Methods

Explainable AI (XAI) and Causal Machine Learning are almost entirely absent from the energy finance literature.

The Paper

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ABSTRACT

Energy finance is an interdisciplinary, rapidly evolving field integrating finance, economics, and energy systems to address complex challenges in market dynamics, sustainability, and policy-making amidst global energy transitions to net zero. Recent advances in artificial intelligence have reshaped energy finance research and practice, but no prior study has systematically examined how these methods have been applied across the field or what value they have brought. This paper provides a comprehensive review of approximately 700 academic studies published over the past five years, analyzing the diffusion and contribution of AI techniques across the six established thematic areas proposed by Zhang et al. (2018). While machine learning models are increasingly adopted, especially in forecasting applications, and often outperform classical econometric approaches in terms of accuracy, their use remains uneven. In some cases, methodological choices are clearly adapted to the structure of the data and the objective of the study, while in others, the implementation of complex ML algorithms lacks contextual justification or interpretability. More importantly, the review highlights a major gap, namely recent approaches such as explainable AI and causal machine learning are almost entirely absent, despite their potential to enhance transparency, support causal inference, and inform energy-related financial decisions. By identifying where AI has already contributed, where it remains underused, and how it could be better integrated, the paper offers a structured and forward-looking perspective on the evolving relationship between artificial intelligence and energy finance.

1. Introduction

Energy is a strategic input to the global economy. Despite international efforts to increase energy efficiency and decouple economic growth from energy consumption, energy demand is rising quickly. In this context, energy products are increasingly viewed as financial assets, leading to a greater interconnection between financial and energy markets. Geopolitical turmoil, particularly the Russian invasion of Ukraine, has intensified the global energy crisis and highlighted the vulnerability of energy systems. Regulatory changes to raise resilience and improve energy risk management, alongside long-term governmental initiatives to enforce more efficient and sustainable resource management, attain net zero, and combat climate change, have further strengthened the link between energy and financial markets. These developments demand a holistic approach toward energy finance.

Much literature emphasizes the interdependency of energy and financial markets, whether presenting energy markets as a source of financial shocks, highlighting the growing financialization of energy

~700 **Articles reviewed**
from Web of Science

6 **Energy finance pillars**
Zhang et al. (2018) framework

5 yrs **Literature coverage**
2020–2025, post-AI explosion

Q1/Q2 **Journal quality filter**
Peer-reviewed only

Methodology: PRISMA 5-Step Approach

Web of Science · Q1 & Q2 journals only

1 Zhang et al. Keywords

2 AI / ML Keywords

3 Pillar-Specific Keywords

4 Journal Quality Filter

5 Relevance Review

Pillar	A: Energy & Fin. Markets	B: Pricing Mech.	C: Corporate Fin.	D: Green Finance	E: Derivatives	F: Risk Mgmt
Identified in WoS	101	103	31+16	231	65	117
Included	40	20	30	48	51	43

Total included: 232 peer-reviewed articles

The Six Pillars of Energy Finance

Framework by Zhang et al. (2018) — structural backbone of this review

A Energy & Financial Markets

B Pricing Mechanisms

C Energy Corporate Finance

D Green Finance & Investment

E Energy Derivative Markets

F Energy Risk Management

Table 3 — Energy Finance Subsections Features

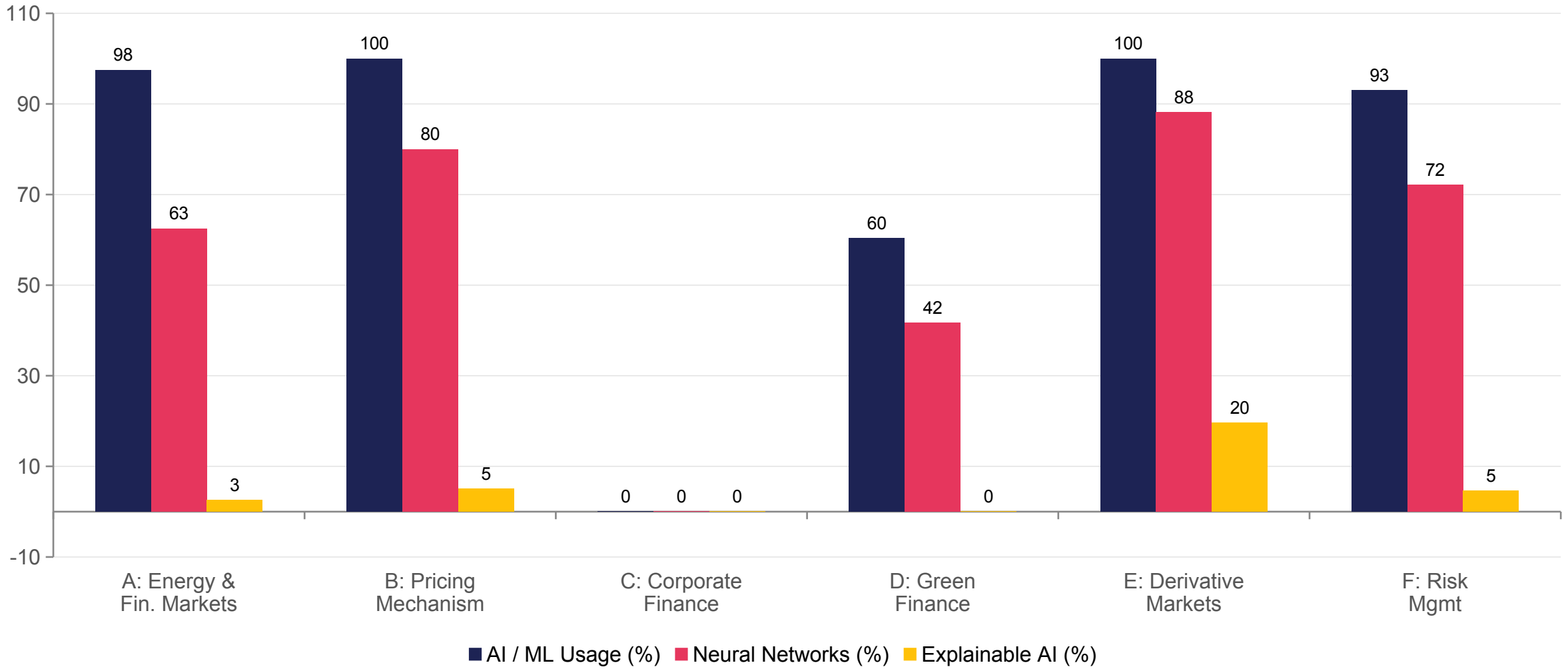
Distribution of AI methods, sectors, geography, techniques across all 6 pillars (from the paper)

Pillar	A: Energy & Fin. Mkts	B: Pricing Mech.	C: Corp. Finance	D: Green Finance	E: Derivatives	F: Risk Mgmt
% Policy papers	72.5%	45%	100%	97.9%	3.9%	27.9%
Target: Oil	62.5%	0%	N/A	31.3%	94.1%	30.2%
Target: Electricity	35%	85%	N/A	85.4%	2.0%	60.5%
EU or G7 focus	75%	45%	36.7%	47.9%	82.4%	67.4%
% Econometrics	100%	25%	100%	58.3%	43.1%	62.8%
% AI/ML	**97.5%**	**100%**	**0%**	**60.4%**	**100%**	**93.0%**
Neural Networks (NN)	62.5%	80%	0%	41.7%	88.2%	72.1%
NLP techniques	7.5%	0%	0%	6.3%	11.8%	2.3%
Explainable AI (XAI)	**2.5%**	**5%**	**0%**	**0%**	**19.6%**	**4.7%**

Source: Table 3 in Anagnoste et al. (2025). XAI row highlighted — the single most underdeveloped dimension across all pillars.

AI Adoption Across the Six Pillars

% of papers using AI/ML, Neural Networks, and Explainable AI



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Pillar A — Energy & Financial Markets

97.5% AI adoption · Focus: oil, clean energy stocks, carbon markets · Daily data

Random Forest (Sadorsky, 2021): 85–90% accuracy predicting clean energy stock direction — vs. 55–60% for logit models

LSTM + Macroeconomic Database (Zhang et al.): Variable selection (LASSO, PCA) across 127 variables improves oil volatility forecasting

Ensemble Methods (Lu et al.): Most consistent performance across conditions for oil futures volatility — combination models add robustness

Sentiment + NLP (Li et al., Loutfi): Text from environmental TV news and media enhances renewable energy stock price forecasting

Bayesian Networks (Wang & Zhao): Captures direct and indirect paths from equity/energy markets to EU ETS carbon futures

Pillar B & C — Pricing Mechanisms and Corporate Finance

One pillar fully transformed by AI · One pillar still AI-free

B Pricing Mechanisms — 100% AI

- Reinforcement learning for battery discharge and storage optimization (Kim et al.)
- K-means clustering segments consumers for personalized demand-response pricing
- BiLSTM + signal decomposition predicts spot electricity prices (Lin et al.)
- ML variable selection refines causal inference on time-of-use pricing in Spain (Enrich et al.)

C Corporate Finance — 0% AI ⚠

- Relies entirely on panel econometrics (GMM, fixed effects)
- Topics: overinvestment, ESG, governance, energy efficiency
- LASSO regression could improve variable selection in sparse panel data
- Random Forests could reveal nonlinear patterns econometrics misses

Pillar D — Green Finance & Investment

60.4% AI adoption · Carbon pricing, renewables, EU ETS, ESG · Mainly Chinese case studies

Key Findings

- Current carbon prices are insufficient to drive significant technological change in polluting industries (Nadirgil, 2023)
- Green finance stimulates renewable investment through fixed asset formation and clean tech innovation
- Financial development improves net-zero transformation potential (ENTP) — but depends on city size and economic structure (Ding et al.)
- EU ETS is a recurring theme: integrating carbon, energy and financial markets since 2005

58%
of papers use AI/ML

62.5%
focus on EU / G7 countries

0%
use XAI — a major gap

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Pillar E — Energy Derivative Markets

100% AI adoption · Focus: oil futures volatility · 51 papers · Daily data dominant

Energy corporate finance	Green Finance and Investment	Energy derivatives market	Energy and risk management
31	231	65	117
0	0	0	0
16	0	0	0
0	0	1	0
0	0	1	0
0	50	6	0
47	171	57	117
17	123	6	74
17	103	6	51
0	20	0	23
30	48	51	43

studies that might have been overlooked by Web of Science queries. Additionally, we otential research directions conducted using machine learning (ML) techniques.

Table 3 organizes these insights by capturing the dominant types of research, the energy sectors most often studied, the geographic focus of the data, and the methodological approaches used. Special attention is given to the adoption of artificial intelligence, not only in terms of overall presence, but also regarding the specific techniques applied, such as neural networks, natural language processing, and other machine learning tools. Moreover, we track the extent to which newer developments like Causality in the classical sense and Explainable AI have been integrated into the literature, particularly in contexts where transparency and policy relevance are essential.

Thus, the first column represents the distribution of various types of research within the literature. For each subfield, we determined the percentage of identified articles that were reviews, benchmarking studies, methodological papers, or policy-oriented papers. This column illustrates the trend in the recent literature on energy finance and the possible gaps that targeted studies could cover. The second element analyzed is related to the energy market sector and presents, for each subfield, the percentage of articles studying the oil, gas, or electricity sector. This column illustrates whether research in a subfield concentrates on specific energy market sectors. The third column refers to the geographical regions. It shows the weight of papers that were carried out based on data for critical economic actors at the global level. For example, it considers the G7, the EU, BRICS countries, and other actors important for the energy market. The forth column presents the percentage of papers that use econometric techniques. The role of this column is to examine whether econometric methods remain the primary vehicle for quantitative analysis in a subfield and whether there are subdomains for which other types of analysis methods are more frequent. The fifth column presents the percentage of AI-oriented papers in each pillar. The purpose is to illustrate the degree of AI utilization. Likewise, the sixth column represents the most common types of AI

Dominant Techniques

🌳 Tree-based methods: Random Forest, GBM — capture nonlinear price dynamics

🧠 LSTM & RNN architectures dominate short-term forecasting — 88% of papers

📊 Decomposition + prediction (EMD, VMD): growing trend for futures volatility

📝 NLP sentiment extraction from news and reports improves oil futures predictions

Pillar F — Energy Risk Management

93% AI adoption · Electricity load forecasting · Renewables · Market volatility

N-BEATS (Oreshkin et al.): Outperforms 10 classical methods in mid-term electricity load forecasting across 35 European countries

Cyber-secure deep learning (Moradzadeh et al.): Autoencoder + CNN + LSTM — maintains accuracy even under false data injection attacks

Reinforcement learning (Zou et al.): Stochastic multi-carrier water and energy management — significant cost reduction

LSTM peer-to-peer trading (Chien et al.): Auto-executable blockchain-based energy market reduces electricity buying costs by 17.68%

Encoder-Decoder RNN (Quiñones et al.): Wind speed and solar radiation forecasting — identifies daily energy surpluses and shortfalls

AI Techniques in Energy Finance

Six methodological families — from forecasting to explanation

Neural Networks (LSTM, CNN, GRU, N-BEATS)

Dominant in ALL pillars — esp. E (88%) & F (72%)

Tree-Based Methods (RF, GBM, XGBoost)

Pillars A & D — oil market, carbon price drivers

Reinforcement Learning (DQN, Policy Gradient)

Pillar B — smart grids, EV charging, battery mgmt

Natural Language Processing (Sentiment, LDA, BERT)

Pillar A — media sentiment, energy stock forecasts

Bayesian / Probabilistic (BNs, Probabilistic Forecast)

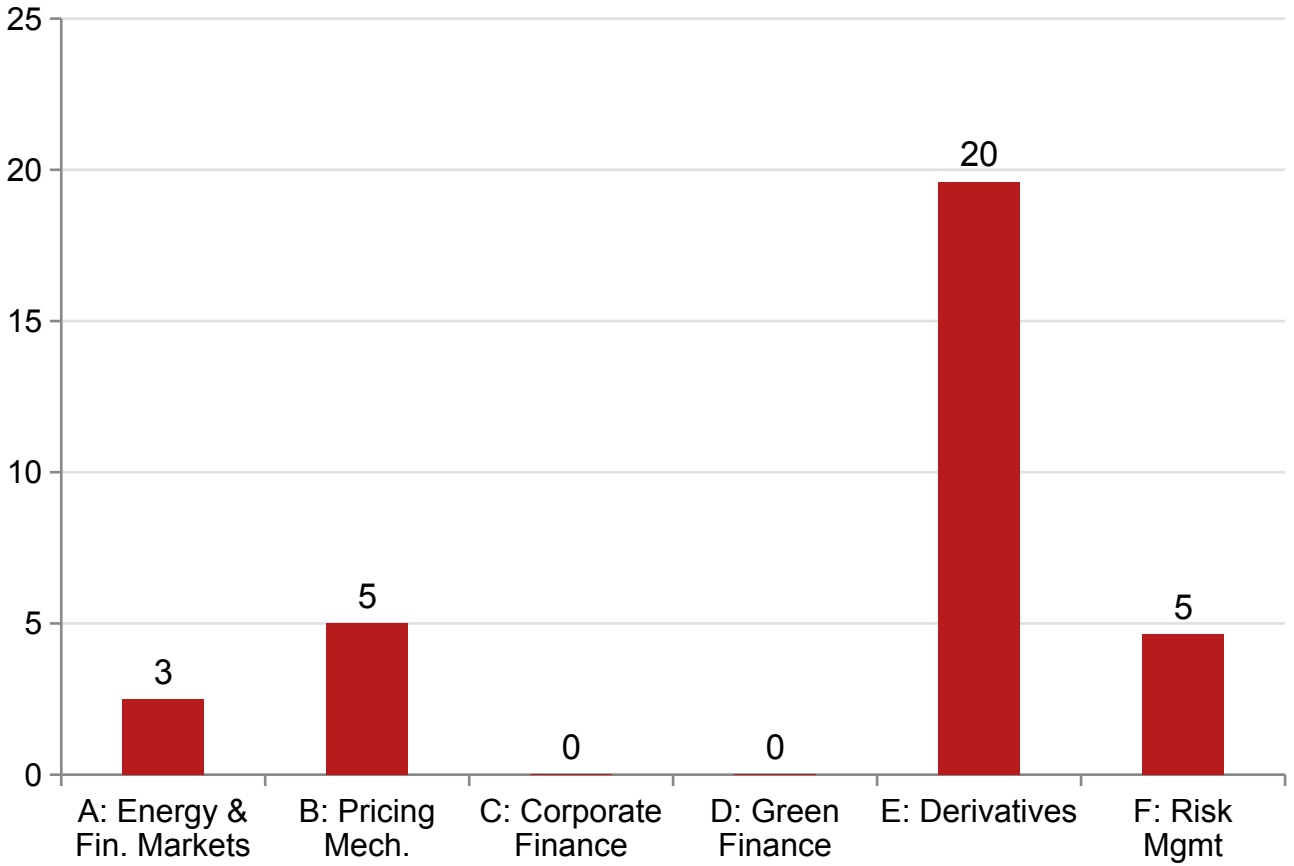
EU ETS carbon price modeling, stress scenarios

Explainable AI + Causal ML (SHAP, LIME, Double ML)

⚠ ALMOST ABSENT — largest research gap

The Critical Gap: XAI & Causal Machine Learning

Almost entirely absent across all 6 pillars — despite clear relevance for policy and transparency



Why It Matters

- Prediction ≠ Causation: ML forecasts prices but cannot explain why an intervention worked
- Policy evaluation requires causal tools — did price caps reduce energy poverty?
- Black-box models cannot justify decisions to regulators or consumers
- ESG behavioral shifts: caused by policy or anticipatory market reactions?

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Energy Finance Quo Vadis — Stakeholder Perspective

Figure 1 from the paper: how pillar connections differ per actor

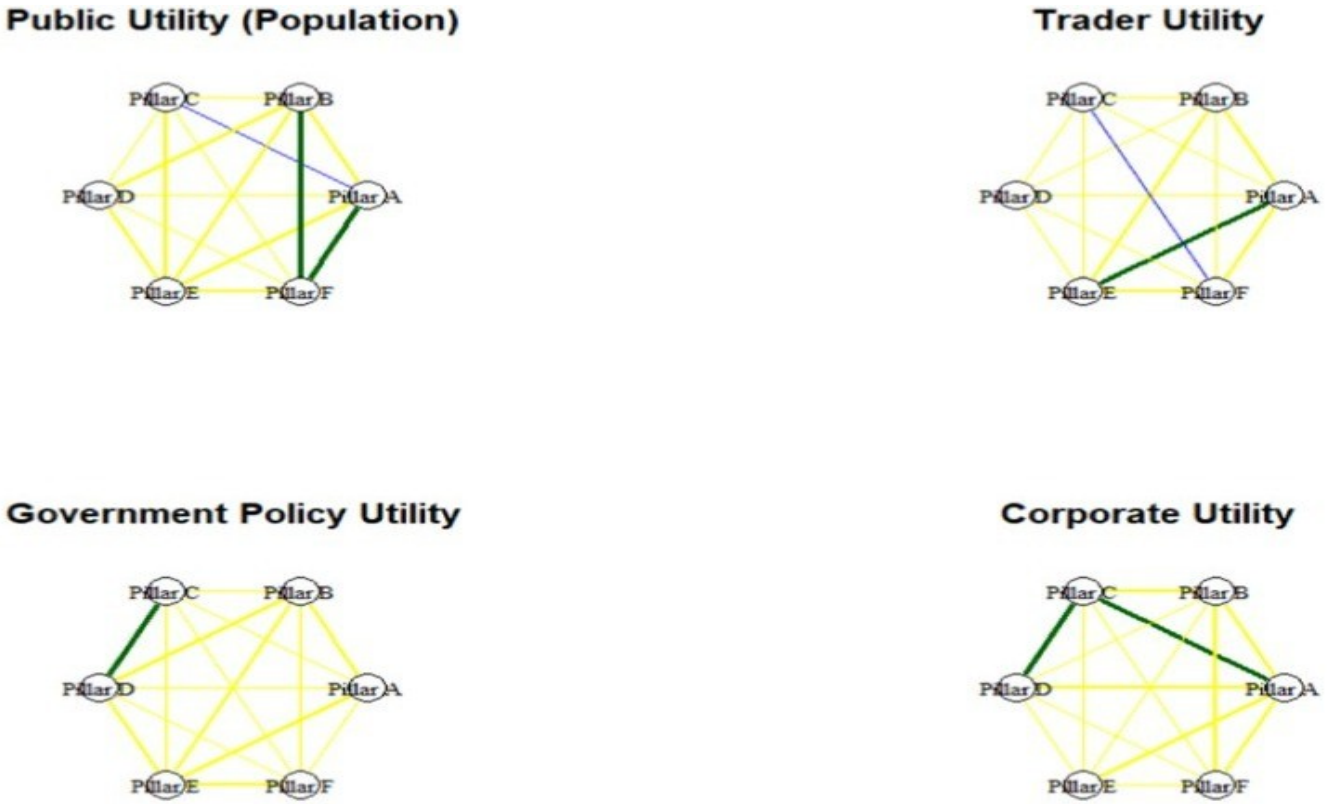


Fig. 1. The connection between pillars based on the utility of each stakeholder
Note: The connections between the pillars shown in the above graphs highlight the degree of interrelation based on common factors such as AI usage, applied methods, energy sectors, and geographic regions relative to the needs of different stakeholder categories (population, traders, governments, and corporations). Dark green lines show strong connections, yellow lines indicate moderate overlap in focus or methods, and blue lines represent weaker links with minimal interaction.
Source: own processing.

Dark green = strong connection · Yellow = moderate overlap · Blue = weak link

What Each Stakeholder Needs from AI

Different actors require very different AI capabilities



Public & Regulators

Need:

Interpretable, transparent models that explain tariff decisions and subsidy allocation

AI tools:

XAI for fairness and consumer trust



Traders & Investors

Need:

Fast, accurate short-term forecasting; volatility and arbitrage signals

AI tools:

LSTM, ensemble methods, NLP sentiment



Energy Companies

Need:

Load forecasting, demand response, ESG alignment, capex evaluation

AI tools:

Causal ML to validate green finance impact



Governments

Need:

Long-term policy evaluation: did interventions achieve intended effects?

AI tools:

Causal ML — double ML, DiD, counterfactuals

A Critical Observation: Methodological Fit

AI adoption should be driven by data structure — not convention

✔ AI Adoption Well-Justified

- High-frequency daily/intraday data in Pillars A, B, E, F
- Complex nonlinear dynamics in oil, electricity and carbon markets
- Shallow NNs, SVMs and tree-based ensembles perform well with typical dataset sizes
- Pattern recognition and re-training are economically valuable for traders

⚠ AI Use Poorly Justified

- Sparse annual panel data in Corporate Finance (Pillar C) — only ~few hundred observations
- Deep learning architectures applied without benchmarking vs. classical methods
- Complex models introduced without explaining contribution to interpretability
- In Green Finance: methodology chosen by fashion, not by data structure

Future Research Agenda

From prediction to explanation — the next frontier

1

Causal ML in Policy Evaluation

Apply Double ML, DiD and counterfactual methods to evaluate energy price caps, green subsidies and environmental taxes — answering did it work, not just what happened next.

2

XAI for Regulatory Transparency

Integrate SHAP/LIME into tariff adjustment models and carbon pricing frameworks. Interpretability improves public trust and supports regulatory compliance.

3

Corporate Finance + ML

Combine LASSO, Random Forests and Causal ML with existing panel data models (FE, GMM) to uncover nonlinear investment behavior in energy companies.

4

Distributional Impact Analysis

Evaluate green subsidies and energy policies across income deciles — did interventions benefit high-income households disproportionately? Micro-level causal analysis needed.

Key Findings at a Glance

Six takeaways for researchers, practitioners and policymakers

01

AI dominates forecasting pillars

A, B, E & F show 93–100% AI adoption. Neural networks and ensembles outperform classical econometrics in price and volatility forecasting.

02

Corporate Finance is AI-free

Pillar C uses 0% AI. Panel econometrics still reigns — but LASSO, RF and Causal ML could meaningfully enrich the field.

03

XAI & Causal ML absent

Despite clear potential for policy evaluation and distributional analysis, explainable AI appears in fewer than 5% of papers across most pillars.

04

Data drives methodology

High-frequency data justifies ML. Sparse annual panel data does not. Some papers apply deep learning without clear rationale.

05

Oil & electricity dominate

Oil leads in A & E; electricity in B & F. Green finance (D) is concentrated in Chinese case studies. EU, G7, BRICS coverage prevails.

06

Next frontier: explanation

The field is moving from prediction to policy insight. XAI + Causal ML are the tools most urgently needed but least deployed.

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Practical Implications

Who benefits and how

Researchers

First structured mapping of AI across all 6 energy finance pillars. Clear identification of where AI is overused without justification — and where it is conspicuously absent.

Traders & Institutional Investors

Ensemble methods and LSTM are proven performers for oil, gas and electricity markets. Explainable components can sharpen risk strategies under volatile regulatory environments.

Energy Companies

AI in load forecasting and demand response delivers measurable cost reduction. Causal ML can validate whether green finance initiatives genuinely affected capital allocation.

Policymakers & Regulators

XAI and Causal ML can reduce information asymmetries, improve allocation of energy resources and assess distributional fairness of energy interventions — tools currently underused.

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- Renewable energy transition, energy security, climate policy
- Quantitative and qualitative methodologies welcome
- Indexed in WoS, Scopus, ScienceDirect

Thank You!

AI4EFin

Roxana Clodnitchi

Corresponding Author
roxana.clodnitchi@fabiz.ase.ro

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linkedin.com/company/ai4efin

github.com/danpele/AI4EFin

www.ai4efin.ase.ro



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